NCDC

NUST CHIP DESIGN CENTRE

Digital Logic Design

Error Detection and Correction

Error Detection and Correction

Part I: Simple Algorithms

- Parity: Single bit detection
- Hamming Codes: Simple error correction

Part II: Detecting Multiple Errors

- Theory: Modulo 2 arithmetic
- Practice Cyclic redundancy codes

The Big Idea: Redundant Encoding

- When storing or sending information, add extra bits (called redundancy)

Error Detection

- When accessing or receiving information, use the extra bits to deduce if corruption occurred.
- Trivial example: send each bit twice, if the values don't match, there was corruption.

Error Correction

- When accessing or receiving information, use the extra bits to fix errors in the data.
- Trivial example: send each bit three times, use a voting scheme to correct single bit errors.
- All error correction has its limits. Trivial example: what if noise corrupted two of the three copies?

Detection Versus Correction

- What are the tradeoffs between detection and correction?

Error Detection

+ Needs fewer redundancy bits.

- If an error is detected, receiver requests a resend of the data. This adds latency and complexity.
- If information was destroyed (like a DRAM flipping a bit) there is no data to resend.

Error Correction

+ Low Latency, correction happens as a part of the logic for receiving the data.

- Uses more redundancy bits.
- If too many errors happened, data cannot be corrected, and a resend will be needed anyways.

Simple Error Detection

Problem: Send an N-bit quantity $b_0 \ldots b_{n-1}$.

Give the receiver a way to detect a single bit error.

<u>Solution: Parity</u>

- Send N+1 bits ($b_0 \ldots b_{n-1}$).
- Bit b_n is one if there is an odd number of ones amongst $b_0 \ldots b_{n-1.}$
- The extra bit is called **parity bit**.

<u>Computing Parity Bit</u>

- $b_n = b_0 \oplus b_1 \oplus \ldots b_{n-1}$
- Where $\oplus$ is the XOR function.
- Parallel data: combinatorial XOR tree.
- Serial data: XOR gate and register bit.

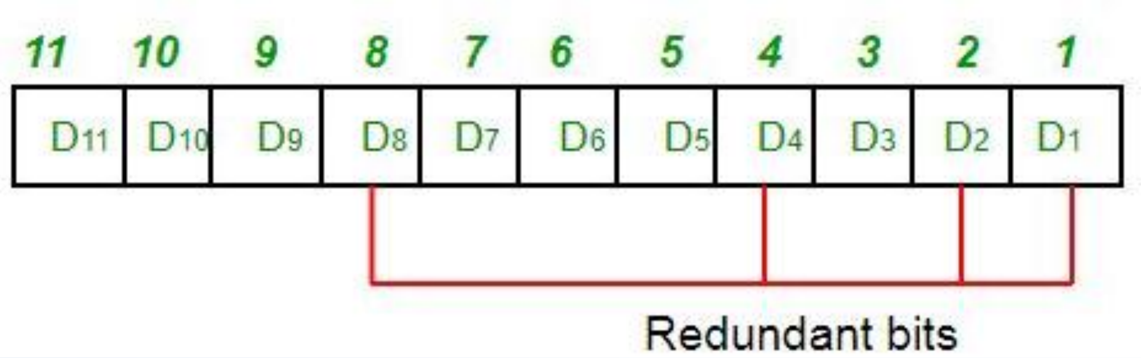
Simple Error Correction

Problem: Send an N-bit quantity $b_0 \ldots b_{n-1}$. Give the receiver a way to detect and correct a single-bit error.

<u>Hamming Codes</u>

- Take overlapping subsets of $b_0 \ldots b_{n-1}$.
- Compute one parity bit for each subset.
- Send *M* bits: *N* data bits and *K* parity bits.
- With enough parity bits, you can ID a flipped data bit.
- Receiver computers a checkword that is 0 (no errors) or whose value is the bit position of the error.
- Complement the bit position of the error to fix.
- R. Hamming, 1950.

Hamming Codes in Detail



<u>Definition</u>

- *N* data bits $b_0 \ldots b_{n-1}$.
- *K* parity bits $p_0 \ldots p_{k-1}$.
- *M total* bits $c_1 \ldots c_M$.
- The c's are numbered starting from **one**.

Step 1: Assign b_i's and p_i's to c_i

- Assign each b_i and p_i only **once**.
- First assign c_1, then c_2, etc.
- Assign c_i to a parity bit if *i* a power of two.
- Elsewise assign c_i to a data bit.
- Stop when no more data bits left to assign.

Assignment Example

Given N – 4: $b_0 \dots b_3$

- $c_1 = p_0$. $(2^0 == 1)$.
- $c_2 = p_1$. $(2^1 == 2)$.
- $c_3 = b_0$.
- $c_4 = p_3$. $(2^2 == 4)$.
- $c_5 = b_1$.
- $c_6 = b_2$.
- $c_7 = b_3$. (last *bi* used)

<u>**Observations**</u>

- $N = 4$, $K = 3$, $M = 4 + 3 = 7$
- For large N, $K \approx log_2(N)$
- This is a good thing.

Bit Position	Binary Index	P0	P1	P3
1 (p_0)	0001	✅	❌	❌
2 (p_1)	0010	❌	✅	❌
3 (b_0)	0011	✅	✅	❌
4 (p_3)	0100	❌	❌	✅
5 (b_1)	0101	✅	❌	✅
6 (b_2)	0110	❌	✅	✅
7 (b_3)	0111	✅	✅	✅

Hamming Codes - Computing Parity Bits

Step 2: Compute Parity bits

- Recall from introduction …
- Take overlapping subsets of the bits.
- Compute one parity bit for each subset.

The Question:

- How to take the subsets from $c_1 \ldots c_M$?

Algorithm:

- Write *c* subscripts in binary.
- Subsets have a 1 in a particular bit position.
- Each subset has only one p_i.
- Set p_i to XOR of subset data bits.

Parity Computation Example

Recall M = 7

- $c_1 = c_{001} = p_0$
- $c_2 = c_{010} = p_1$
- $c_3 = c_{011} = b_0$
- $c_4 = c_{100} = p_3$
- $c_5 = c_{101} = b_1$
- $c_6 = c_{110} = b_2$
- $c_7 = c_{111} = b_3$

Bit Position	Binary Index	P0	P1	P3
1 (p_0)	0001	✅	❌	❌
2 (p_1)	0010	❌	✅	❌
3 (b_0)	0011	✅	✅	❌
4 (p_3)	0100	❌	❌	✅
5 (b_1)	0101	✅	❌	✅
6 (b_2)	0110	❌	✅	✅
7 (b_3)	0111	✅	✅	✅

Subsets

- $(c_4, c_5, c_6, c_7) == (p_3, b_1, b_2, b_3)$
- $(c_2, c_3, c_6, c_7) == (p_1, b_0, b_2, b_3)$
- $(c_1, c_3, c_5, c_7) == (p_0, b_0, b_1, b_3)$

To Compute Parity

- XOR data bits of the subset.
- Assign to parity bit of the subset.
- Each subset has one parity bit by design.

Hamming Codes – Correcting Error

<u>**Step 3: Correct Errors on Reception**</u>

- With enough parity bits, you can ID a flipped data bit.
- Receiver computes a checkword that is 0 (no errors) or whose value is bit position of the error.

The Question:

- How to compute checkword?

Algorithm:

- Break data into subsets.
- If subset contains p_i, set w_i to XOR of all subset members.
- Checkword = $w_0 + 2 * w_1 + 4 * w_2 + \ldots$
- If checkword is zero, no errors.
- If checkword is non-zero, flip bit $c_{checkword}$.

Checkword Computation Example

$c_1 = c_{001} = p_0$

$c_2 = c_{010} = p_1$

$c_3 = c_{011} = b_0$

$c_4 = c_{100} = p_3$

$c_5 = c_{101} = b_1$

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$c_7 = c_{111} = b_3$

Subsets

$(c_4, c_5, c_6, c_7) == (p_3, b_1, b_2, b_3)$

$(c_2, c_3, c_6, c_7) == (p_1, b_0, b_2, b_3)$

$(c_1, c_3, c_5, c_7) == (p_0, b_0, b_1, b_3)$

To Compute Checkword

$w_2 = c_4 \oplus c_5 \oplus c_6 \oplus c_7 = p_3 \oplus b_1 \oplus b_2 \oplus b_3$

$w_1 = c_2 \oplus c_3 \oplus c_6 \oplus c_7 = p_1 \oplus b_0 \oplus b_2 \oplus b_3$

$w_0 = c_1 \oplus c_3 \oplus c_5 \oplus c_7 = p_0 \oplus b_0 \oplus b_1 \oplus b_3$

Checkword = $w_0 + 2 * w_1 + 4 * w_2$

Bit Position	Binary Index	P0	P1	P3
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4 (p_3)	0100	❌	❌	✅
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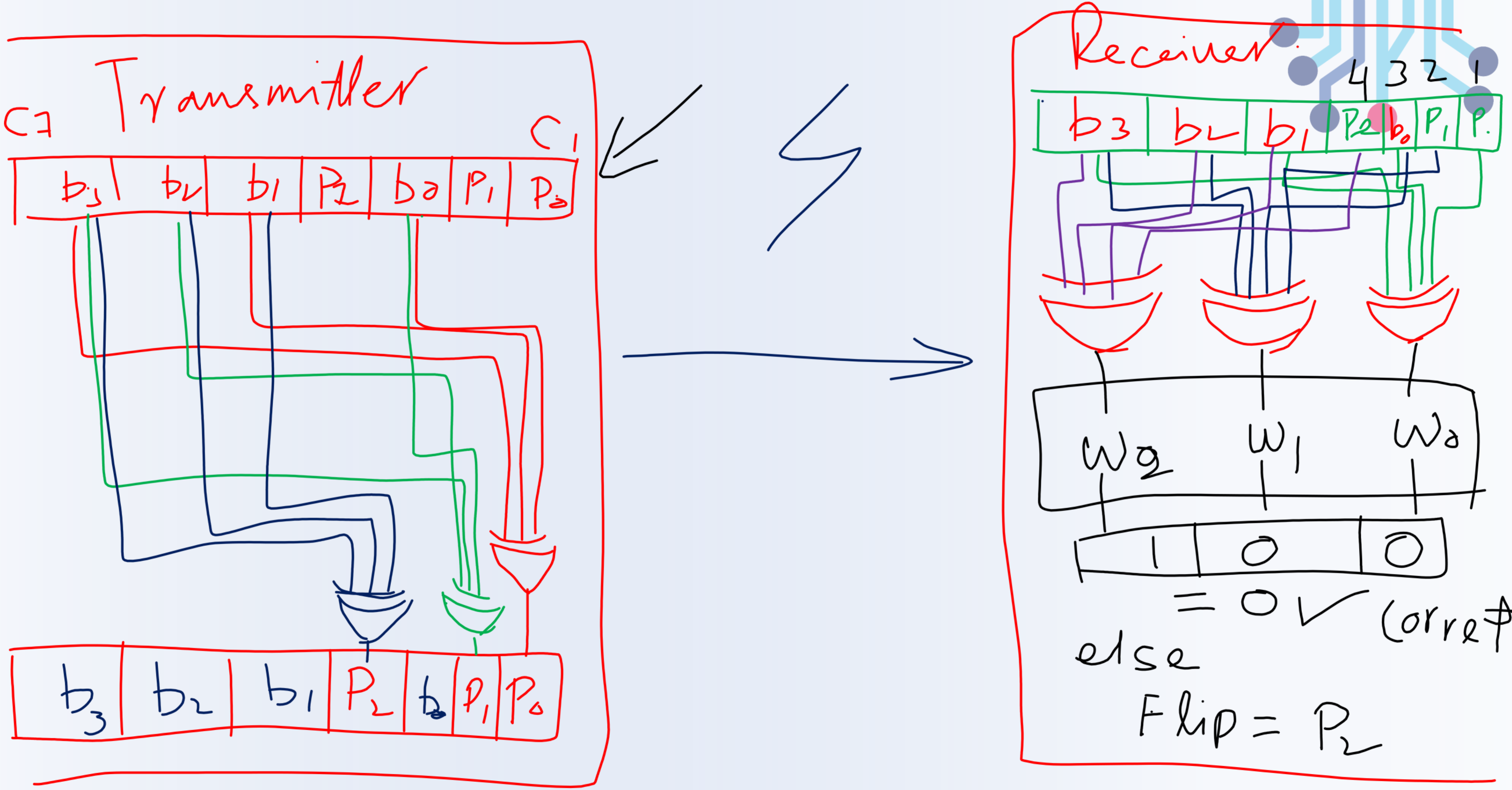


C7
Transmitter
C1
b3 b2 b1 P2 b0 P1 P0
b3 b2 b1 P2 b0 P1 P0
Receiver
4 3 2 1
b3 b2 b1 P2 b0 P1 P0
W2 W1 W0
1 0 0
= 0 ✓ correct
else
Flip = P2

Hamming Code Epilogue

Why Does it Work?

- Uses binary code to advantage.
- XOR all subsets which will result in binary number.
- If checkword is zero, no error else a bit must have changed.
- The number of checkword indicate the position of flipped bit.
- Just invert the bit at the checkword location and message is corrected.

Activity

Identify the corrupted transmissions and rectify the error.

	P_0	P_1	B_0	P_2	B_1	B_2	B_3	P_3	B_4	B_5	B_6	B_7
Bit position	1	2	3	4	5	6	7	8	9	10	11	12
	0	0	1	1	1	0	0	1	0	1	0	0
	1	0	1	1	1	0	0	1	0	1	0	0
	0	0	1	1	0	0	0	1	0	1	0	0

Detecting Multiple Errors

For Use When:

- Sending packets of data over an unreliable medium.
- Most of the time, bits arrive perfectly.
- Sometimes, many bits are corrupted.
- Need to detect corruption, request a resend.
- Parity too weak: multi-bit corruption is common.

Example Applications:

- Ethernet Packets
- Internet Protocol Datagrams

Basic Idea: Checksums

- Given a data packet of *N* bits.
- Compute an *M* bit **checksum** word on data.
- Send *M* checksum bits and *N* data bits.
- Receiver also computes checksum on packet.
- Parity is a very weak 1-bit checksum.

Desirable Checksum Properties

- *N* may vary packet by packet, *M* is fixed.
- Algorithm should be easy to compute.
- Algorithm and *M* chosen to match application.

Example: Cyclic Redundancy Check (CRC)

- Used in Ethernet Packets ($M = 32$).
- Ethernet P (undetected error) = $1/2^{32}$
- Less than one in a billion.

The Basic Idea

- Treat data packet as an N bit number.
- Divide number by a constant.
- The “remainder” of the division is the checksum.

But Division is Slow!

- Solution: use a number system where division is fast.
- Modulo 2 Arithmetic.

CRC Method

CRC is a method for detecting errors in digital data (single-bit, burst errors, etc.).

How does it work:

- Choosing a **generator polynomial** G (pre-agreed between sender and receiver).
- Treating the data bits as a binary polynomial D.
- Appending **r** zeros to D (where r = degree of G).
- Doing **modulo-2 division** of $D \times 2^r$ by G to get the **remainder** R.
- Sending **{data, remainder}**.
- Receiver repeats the same division with {data, remainder}, if remainder = 0, no error is assumed.

Modulo 2 Arithmetic

In **modulo-2 arithmetic**:

- Addition = XOR
- Subtraction = XOR (same as addition)
- No carries/borrows → simple logic
- Division just means:
  - Align generator with current dividend bits
  - If MSB is 1, XOR with generator; shift; repeat
  - If MSB is 0, just shift
- Because everything is XOR and shifts, CRC is super cheap in hardware — just shift registers + XOR gates.

Modulo 2 Division

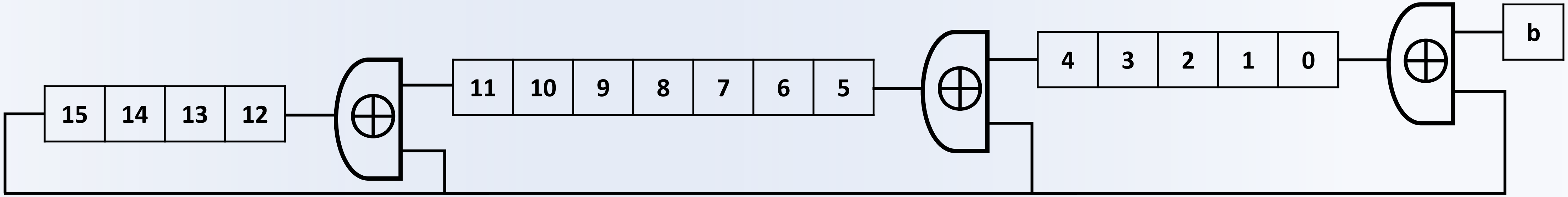
```
                    1 0 1 0 1 1
            ___________________
    1 0 0 1 | 1 0 1 1 1 0 0 0 0
              1 0 0 1
              -------
                  1 0 1
                  0 0 0
                  -----
                  1 0 1 0
                  1 0 0 1
                  -------
                    1 1 0
                    0 0 0
                    -----
                    1 1 0 0
                    1 0 0 1
                    -------
                      1 0 1 0
                      1 0 0 1
                      -------
                        0 1 1
```

Generator

Data

Remainder

CRC Hardware



- Serial dividend flows from *b* box.
- Flip Flops hold 16-bit remainder *R*.
- 17-bit divisor *G* = 10001000000100001
- Input XOR is first and last 1.
- Two middle 1's map to middle XORs.

Preparing Checksum

- Clear flip-flops.
- Append 16 zeros to *N*-bit data *D*.
- 16 + *N* clocks to enter data.
- Remainder *R* sits in flip-flops.

Upon Receipt of $2^{16} * D + R$

- Clear flip-flops.
- 16 + *N* clocks to enter data.
- A non-zero flip-flop flags an error.

CRC – How does it Work?

- The Secret: it doesn’t work for all divisors.
- Only for numbers with “cyclic” property.
- Theoretical background needed to go further.
- Linear Feedback Shift Registers
- Also used for random number generation.

In Conclusion

- Error coding is where match meets logic.
- Modulo 2 bit-serial circuits are powerful.
- Deep mathematics behind the gates.

Thank You

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